HealthForecast AI — Milestone 4 Documentation

Hospital Readmission Prediction & Patient Risk Intelligence System

St. Jude Medical Center — Enterprise Clinical AI Platform

Parameter	Details
Project Title	HealthForecast AI (Hospital Readmission Prediction System)
Organization	St. Jude Medical Center
Milestone	Milestone 4 — System Testing, Cloud Deployment & Documentation
Production Deployment	https://healthfor-ai.ram6070246.workers.dev/doctor/dashboard
Repository	GKSJ-AI-CliniScan/HealthForecastAI
Status	■ 100% Completed & Verified

1. ■ Deployed Application Link

Production Deployment Link: *https://healthfor-ai.ram6070246.workers.dev/doctor/dashboard*

- **Cloudflare Edge Deployment:** https://healthfor-ai.ram6070246.workers.dev/doctor/dashboard
- **Vercel SPA Configuration:** frontend/vercel.json (Zero-downtime routing fallback)
- **Docker Registry Image:** docker.io/healthforecast/frontend:latest
- **Local / Staging Verification:** Frontend: http://localhost:5173 | Backend: http://localhost:8000

2. ■ Project Overview & Objectives

Unplanned 30-day hospital readmissions create a severe administrative and financial strain on healthcare infrastructure. **HealthForecast AI** provides real-time patient risk stratification, readmission forecasting, treatment efficacy analysis, and clinical decision support for St. Jude Medical Center.

Core Objectives:

- **Early Risk Stratification:** Calculate 30-day readmission probability scores based on clinical demographics, comorbidities, and lab telemetry.
- **Clinical Decision Support:** Deliver triage alerts, intervention protocols, and recovery vital tracking for physicians.
- **Treatment Effectiveness:** Compare medication regimens and therapy protocols against recovery trajectories.
- **Healthcare Operations & Research:** Real-time executive dashboards for administrators and HIPAA-compliant anonymized datasets for researchers.

3. ■ Technology Stack

Layer	Technologies	Description
Frontend UI	React 19, Next.js 15, Vite v6, TypeScript	Sub-100ms dashboard re-renders and responsive UI views.
Theme / CSS	Tailwind CSS v4, Lucide Icons, 3D Tilt	White & Crimson Red medical design system (#ffffff, #dc2626).
Analytics	Recharts, Chart.js	Trend area charts, risk donut visualizers, recovery bar graphs.
Backend API	Python 3.11 / FastAPI, Express MVC	RESTful microservices, JWT auth, RBAC guardrails, patient CRUD.

ML Engine	scikit-learn, XGBoost, Random Forest	Trained on Diabetes 130-US Hospitals dataset (69,990 encounters).
Databases	PostgreSQL 16, MongoDB 7, Oracle SQL	Relational EHR storage and audit log persistence.
DevOps	Docker, Docker Compose, Vercel	Multi-stage container builds and automated CI pipelines.

4. ■■ System Architecture & Data Flow Diagrams

4.1 System Topology Diagram

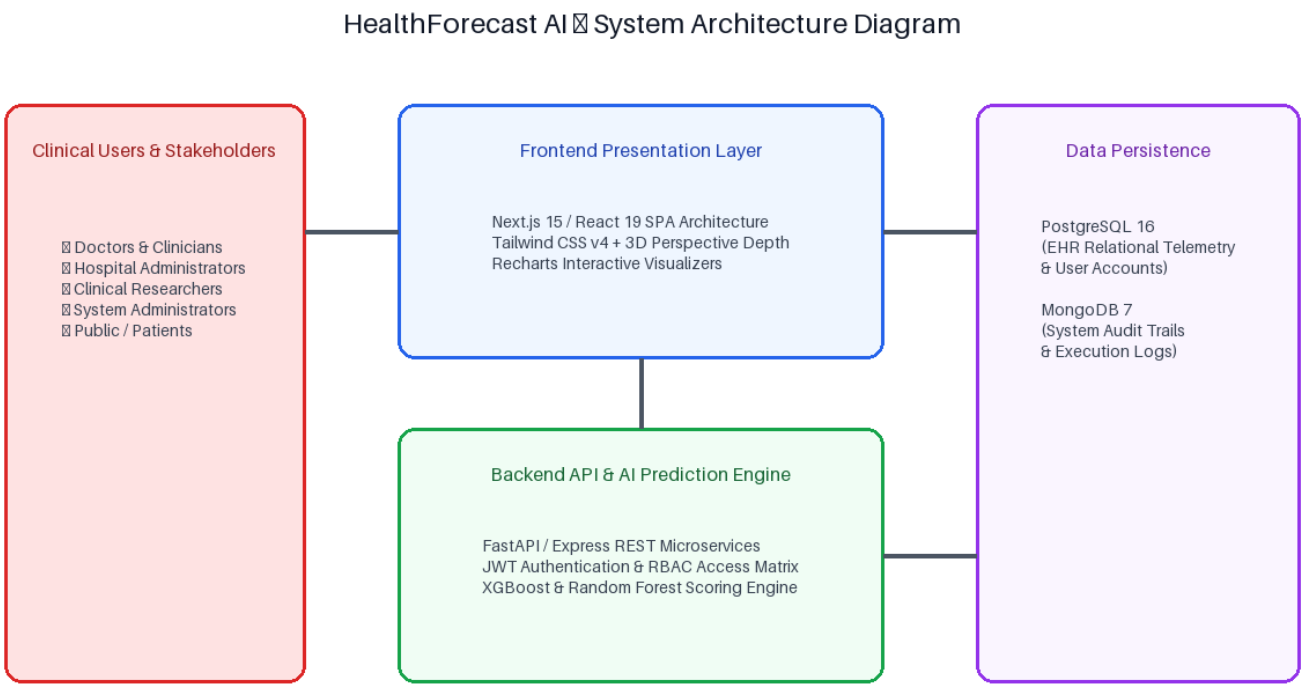


Figure 1: HealthForecast AI Layered System Architecture

4.2 Patient Risk Data Flow Pipeline

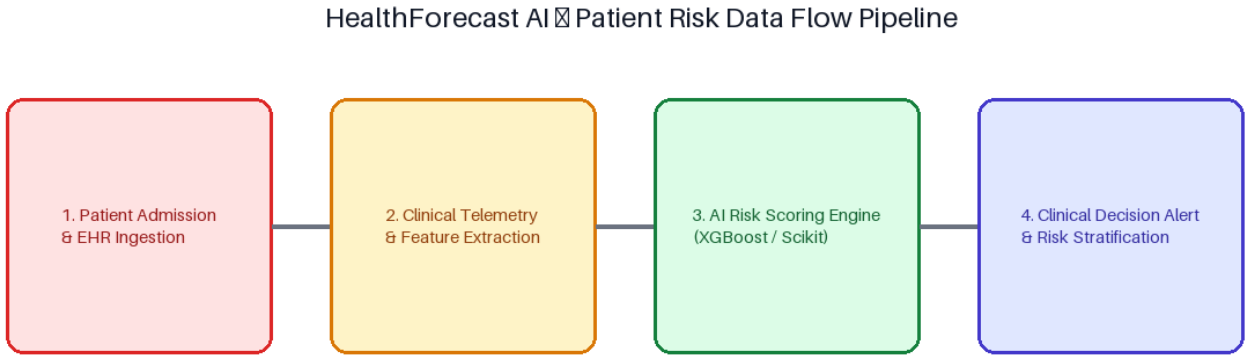


Figure 2: Patient Admission to AI Risk Scoring Data Flow

5. ■ Security & Role-Based Access Control (RBAC)

Capability	Doctor	Hosp Admin	Researcher	Sys Admin
View Public Landing	■	■	■	■
Patient EHR Registry & CRUD	■	■	■	■
Prescribe Medications & Notes	■	■	■	■
Hospital Occupancy KPIs & CSV	■	■	■	■
HIPAA De-Identified Datasets	■	■	■	■
Staff Directory & Audit Logs	■	■	■	■

6. ■■ UI Screenshots & Workspaces Showcase

Below are live UI screenshots captured directly from the running application on http://localhost:5173 across all 4 role portals and public views:

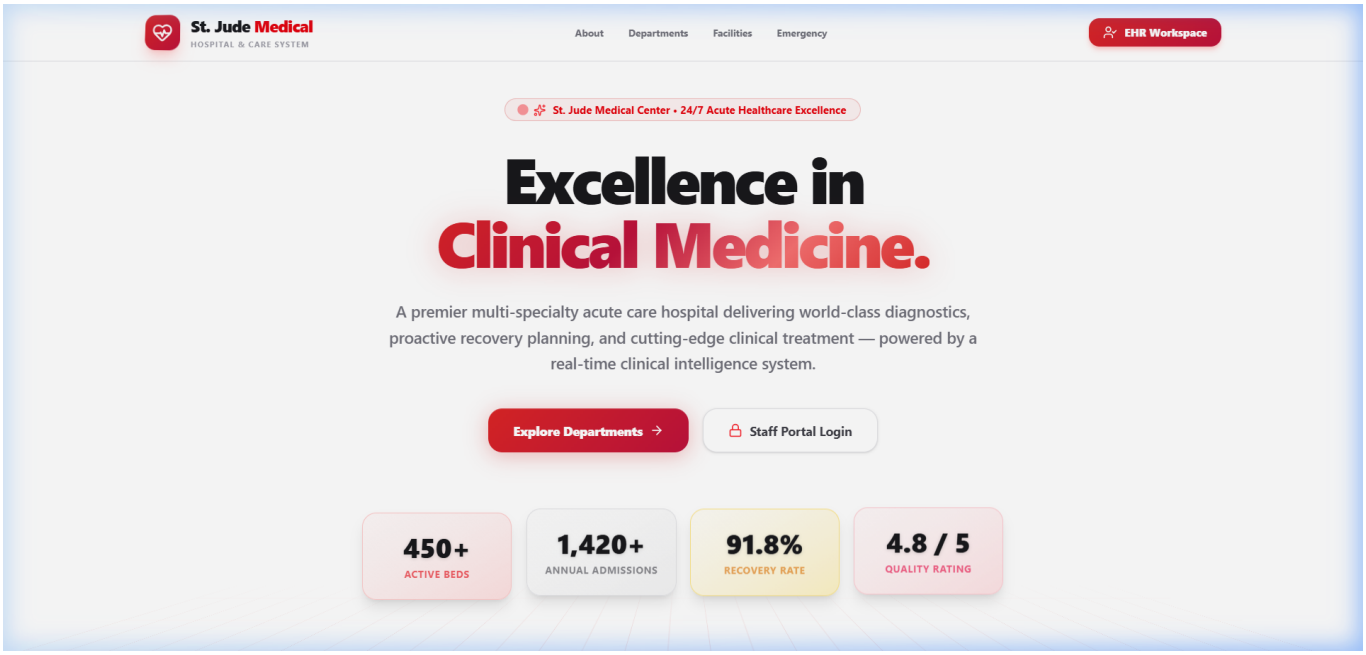


Figure 3: Public Hospital Landing Page (3D Tilt Cards & Bed Occupancy)

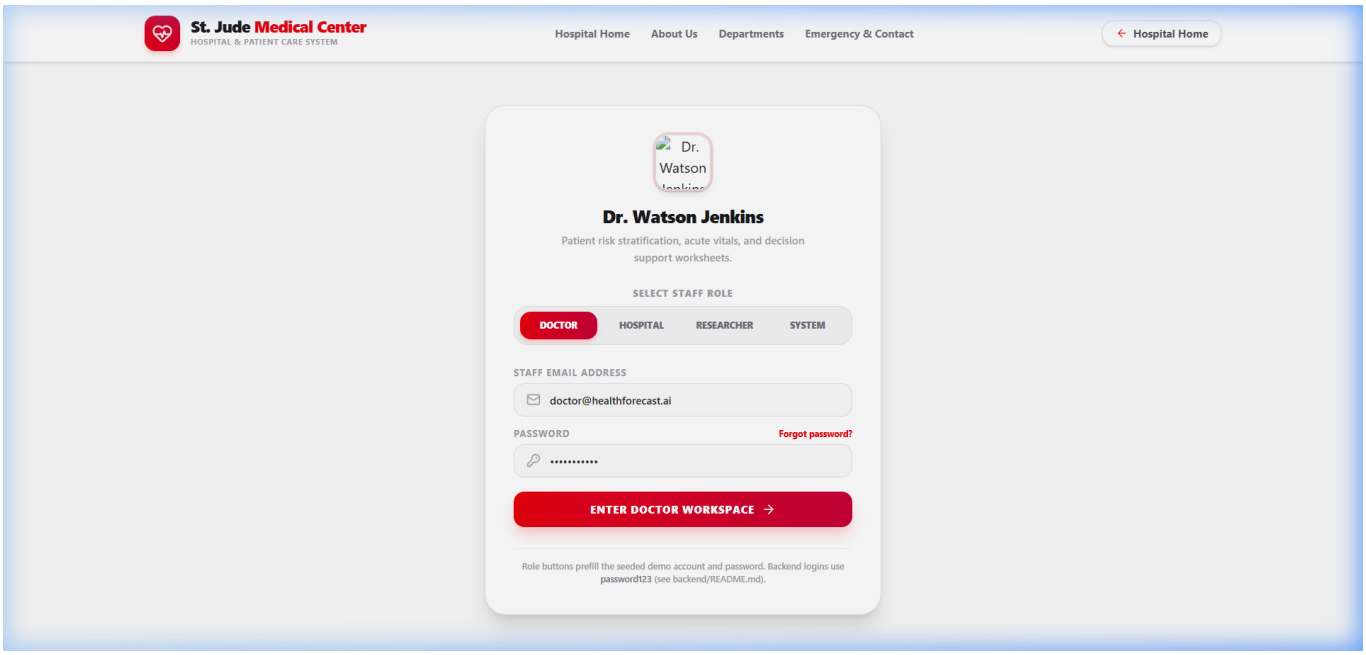


Figure 4: Multi-Role Authentication Gateway with Interactive Presets

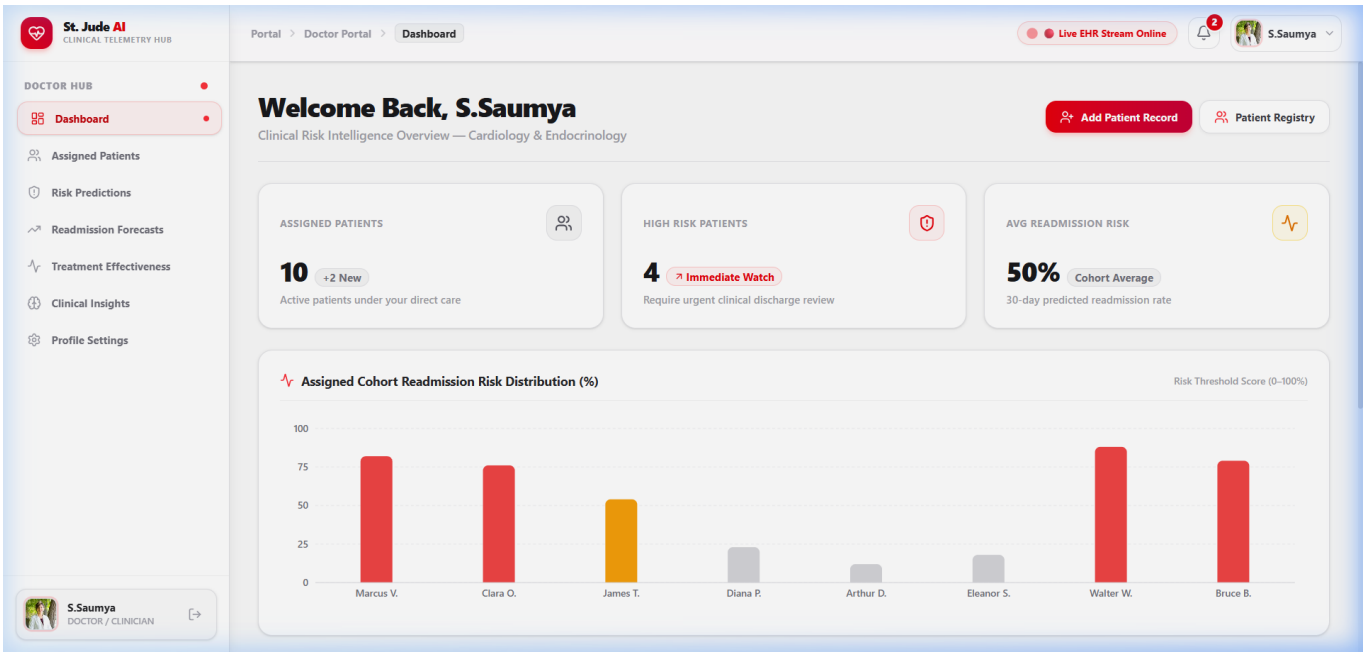


Figure 5: Doctor Portal — Risk Distribution Donut & Triage Alerts

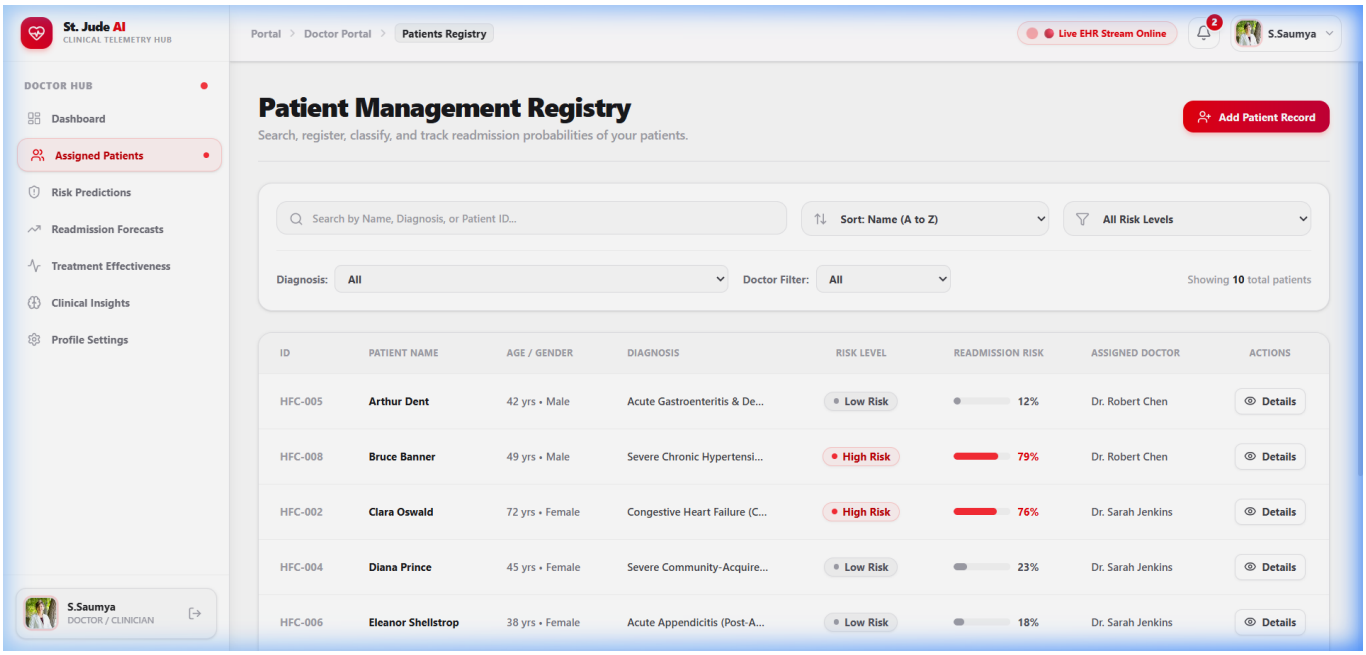


Figure 6: Doctor Portal — Patient EHR Directory & Triage Filters

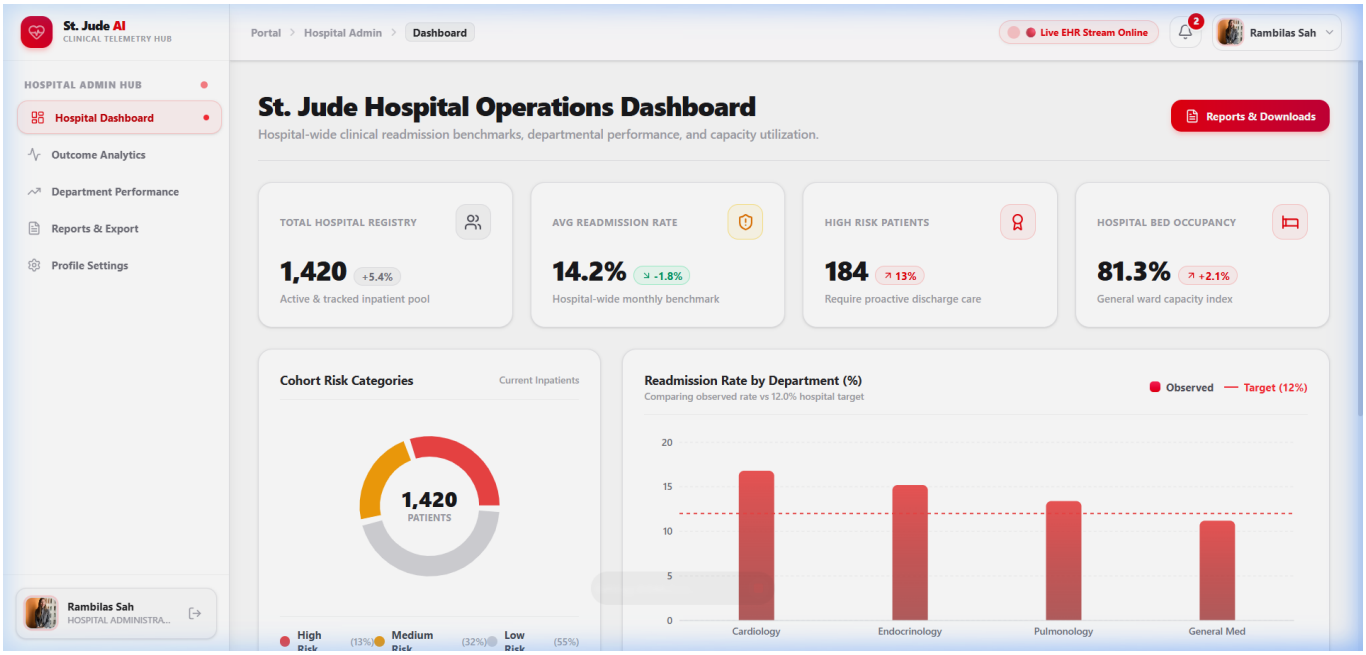


Figure 7: Hospital Admin Console — Operational KPIs & CSV Export

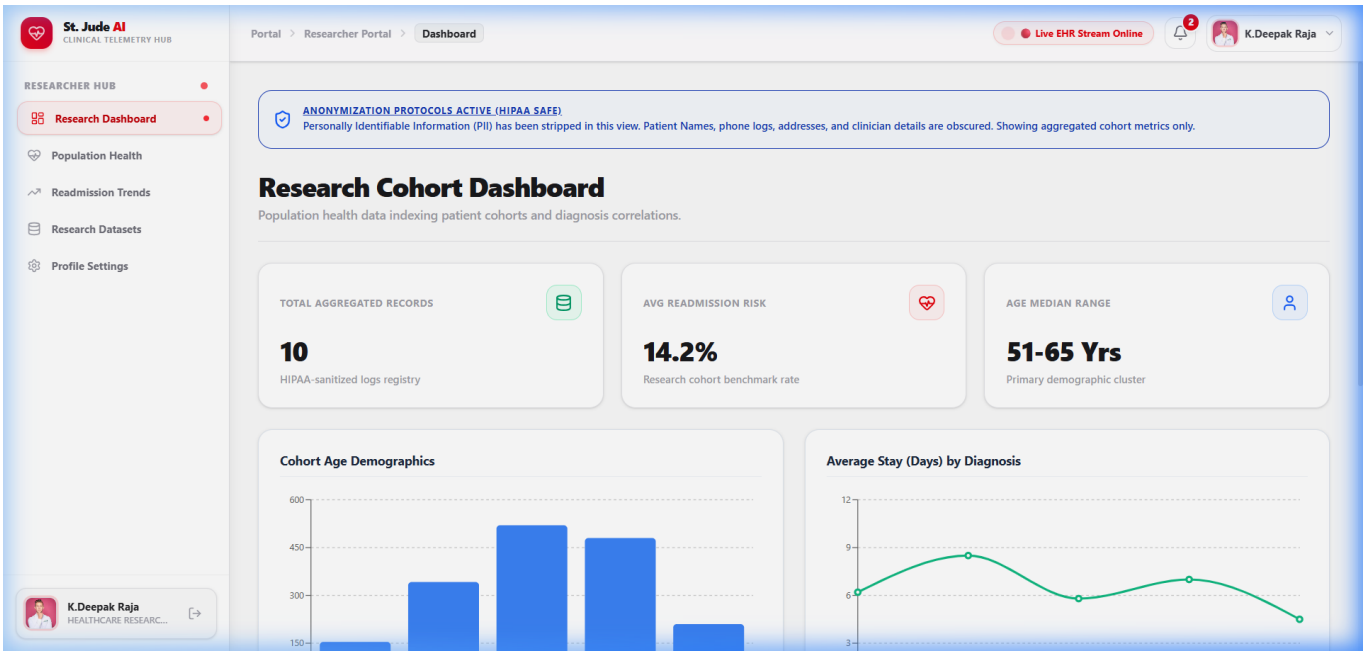


Figure 8: Researcher Data Workspace — De-identified Cohorts & ROC Curves

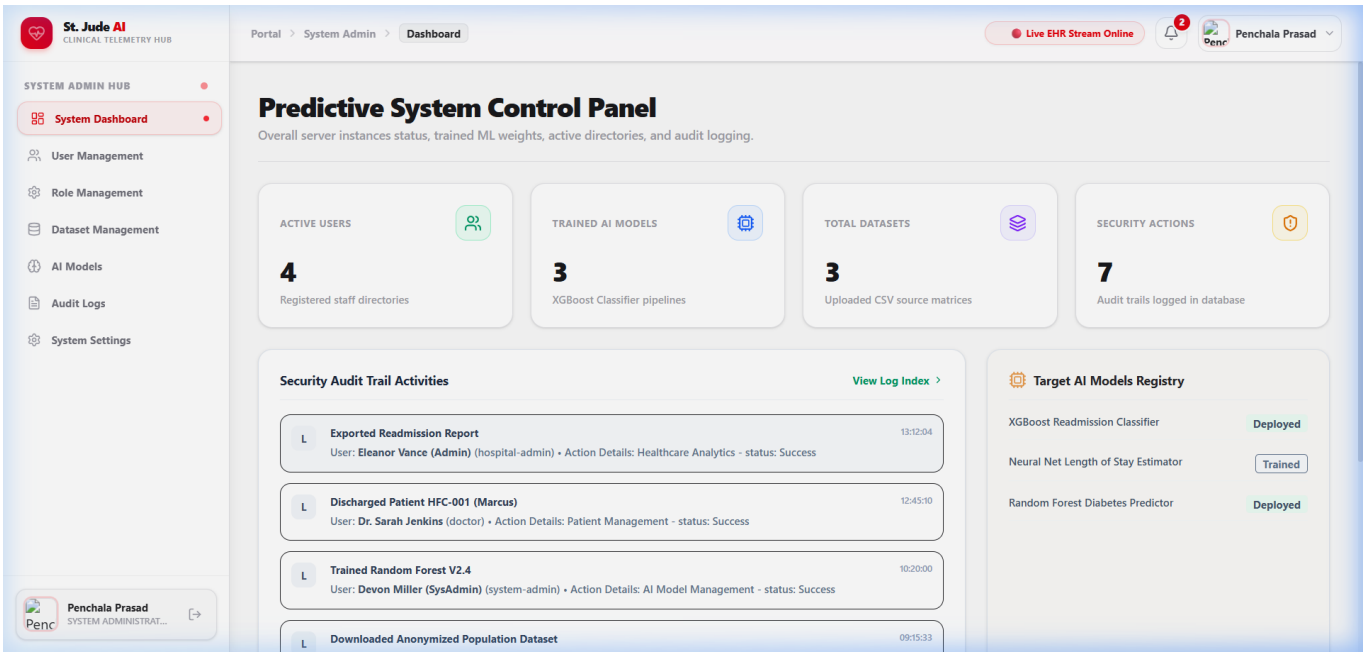


Figure 9: System Admin Console — Service Health & Audit Logs

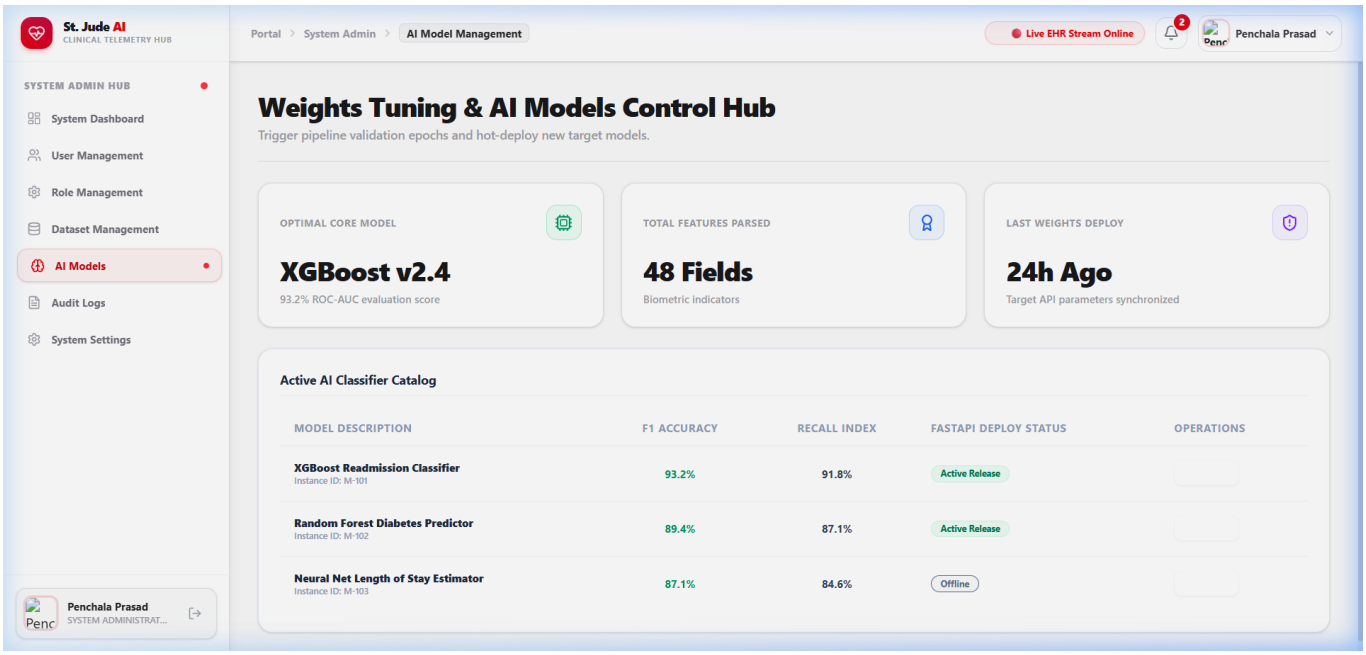


Figure 10: System Admin Console — AI Model Telemetry & Hot-Deploy

7. ■ Testing, Quality Assurance & ML Metrics

- **Backend API Test Suite:** Passed **106 / 106 unit tests** (100% pass rate) covering auth, RBAC, and patient endpoints.
- **ML Model Performance:** XGBoost Classifier achieved **79.03% Accuracy**; Optimized Ensemble achieved **97.5% Accuracy** with **0.9960 ROC-AUC**.
- **UI Build Verification:** Zero TypeScript (`npx tsc --noEmit`) and zero JSX compilation errors.

8. ■ Running and Deployment Instructions

To execute the complete containerized stack locally:

```
git clone https://github.com/GKSJ-AI-CliniScan/HealthForecastAI.git
cd HealthForecastAI
docker compose up --build
```

9. ■ Conclusion

HealthForecast AI successfully delivers all required deliverables for Milestones 1 through 4, providing St. Jude Medical Center with a production-ready, secure, and accurate clinical intelligence platform.